

Deploy Hortonworks Sandbox on Google Cloud:

The following steps need to be done only for the first time:

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1. Login to your Google Cloud account (<https://console.cloud.google.com/>) and open a Cloud Shell. Make sure you are in the intended project.
2. Run the following command in **Cloud Shell** to launch a new instance named ambari. Copy it into an editor of your choice (Shell Editor is preferred) and replace <PROJECT-ID> with your project ID (click the project name at the top of the console; the ID is listed next to it):

```
gcloud compute instances create ambari \
--project=<PROJECT-ID> \
--zone=us-east1-b --machine-type=e2-highmem-4 \
--subnet=default --network-tier=PREMIUM --no-restart-on-failure \
--maintenance-policy=MIGRATE \
--image-family=cos-129-lts --image-project=cos-cloud \
--boot-disk-size=50GB --boot-disk-type=pd-balanced \
--boot-disk-device-name=ambari
```

- This gives you a machine with a 50 GB disk and 32 GB of memory (e2-highmem-4). It costs about €18/hour (as of Sep 2026).
 - If you get a ZONE_RESOURCE_POOL_EXHAUSTED error, the zone is temporarily full. Run the command again with --zone=us-east1-c or --zone=us-east1-d, and use that zone wherever us-east1-b appears in the rest of these steps.
3. Once the instance is ready (it has an external IP address), a green checkmark appears next to its name under Compute Engine. Click the SSH button to SSH into your machine.
 - The first time, the SSH button may take a while to work. If so, run this from Cloud Shell instead:

```
gcloud compute ssh ambari --project=<PROJECT-ID> --zone=us-east1-b
```

4. Log in as the root user: `sudo su - root`
5. From the **SSH window**, download the HDP 2.5 image to your instance (~1 minute):

```
wget https://storage.googleapis.com/is843/HDP_2.5_docker.tar.gz -P /var/tmp/
```

6. Load the image into Docker (~5 minutes):


```
docker load < /var/tmp/HDP_2.5_docker.tar.gz
```
7. Start the container with the following command:

```
docker run -v hadoop:/hadoop --name sandbox --hostname \
"sandbox.hortonworks.com" --privileged -d -p 6080:6080 -p 9090:9090 \
-p 9000:9000 -p 8000:8000 -p 8020:8020 -p 42111:42111 -p 10500:10500 \
-p 16030:16030 -p 8042:8042 -p 8040:8040 -p 2100:2100 -p 4200:4200 \
-p 4040:4040 -p 8050:8050 -p 9996:9996 -p 9995:9995 -p 8080:8080 -p 8088:8088 \
-p 8886:8886 -p 8889:8889 -p 8443:8443 -p 8744:8744 -p 8888:8888 -p 8188:8188 \
-p 8983:8983 -p 1000:1000 -p 1100:1100 -p 11000:11000 -p 10001:10001 \
-p 15000:15000 -p 10000:10000 -p 8993:8993 -p 1988:1988 -p 5007:5007 \
-p 50070:50070 -p 19888:19888 -p 16010:16010 -p 50111:50111 -p 50075:50075 \
-p 50095:50095 -p 18080:18080 -p 60000:60000 -p 8090:8090 -p 8091:8091 \
-p 8005:8005 -p 8086:8086 -p 8082:8082 -p 60080:60080 -p 8765:8765 \
-p 5011:5011 -p 6001:6001 -p 6003:6003 -p 6008:6008 -p 1220:1220 \
-p 21000:21000 -p 6188:6188 -p 61888:61888 -p 2181:2181 -p 2222:22 \
```

```
sandbox /usr/sbin/sshd -D
```

- The container should be running now. Confirm with `docker ps`; it should list a running container named `sandbox`.
 - Once the container is running, delete the downloaded file to free up 11 GB of disk:
`rm /var/tmp/HDP_2.5_docker.tar.gz`
8. Start the sandbox services. Ambari and HDP do not start on their own when the container starts, so you need to run a script:
 - Open a shell inside the “sandbox” container we just created:
`docker exec -it sandbox bash`
 - Your prompt changes to `[root@sandbox /]#`, which means you are inside the container. You may see “sed: can't read .bash_logout: No such file or directory”; ignore it.
 - Start the services (~3 minutes):
`/etc/init.d/startup_script start`
 9. While you are here, set a password for the Ambari “admin” account:
`ambari-admin-password-reset`
 - This restarts the Ambari server, so port 8080 will be unavailable for about a minute.
 10. When all services show OK, the Ambari setup is complete. If you are in the SSH window, you can close it now. If you are in Cloud Shell, type `exit` three times (container, root, VM) to get back to the Cloud Shell prompt.
 11. From **Cloud Shell**, create a new firewall rule that allows INGRESS tcp:8080 with VMs containing the tag “allow-tcp-8080”:
`gcloud compute firewall-rules create rule-allow-tcp-8080 \`
`--source-ranges 0.0.0.0/0 --target-tags allow-tcp-8080 --allow tcp:8080`
 - This needs to be done once per project. If you have done it before, you can ignore it.
 12. Set the default zone to the one your instance is in:
`gcloud config set compute/zone us-east1-b`
 13. Add the “allow-tcp-8080” tag to your new VM:
`gcloud compute instances add-tags ambari --tags allow-tcp-8080`

Important note (optional): Steps 11-13 expose port 8080 of the instance to the internet. That is convenient but not secure. A safer option is an SSH tunnel:

- First, Google Cloud SDK has to be installed and initialized:
<https://cloud.google.com/sdk/docs/downloads-interactive>
 - If you are a Windows user and the above step didn't help you can find more info here:
<https://cloud.google.com/sdk/docs/quickstart-windows>
 - Use this command to forward port 8080 and access the instance securely: `gcloud compute ssh ambari --zone us-east1-b -- -L 8080:localhost:8080`
 - From the browser go to `localhost:8080`
14. From the Compute Engine page, write down the external IP address of the “ambari” instance. Then open Ambari in your local browser:
 - `<External-IP-Address>:8080`
 - Log in with `maria_dev` as both the username and password.
 15. **IMPORTANT:** Stop the instance at the end of each session. It costs €18/hour while running, and if you forget to stop it, your \$50 credit will last a little over a week.



Restarting the Stopped Instance

1. From Compute Engine, select your “ambari” instance, and from the top menu, click Start.
2. SSH into the machine (step 3 above) and log in as root:
 - a. `sudo su - root`
3. Restart the container:
`docker restart sandbox`
4. Start the sandbox services. They do not start on their own when the container restarts:
 - a. Open a shell inside the container:
`docker exec -it sandbox bash`
 - b. Run the startup script (~3 minutes):
`/etc/init.d/startup_script start`
5. The external IP address usually changes when the instance is stopped. Look up the current one in Compute Engine and log in to Ambari from your local browser:
 - a. `<External-IP-Address>:8080`

Resetting password for maria_dev

1. Log in to Ambari as “admin” with the password you set in step 9.
2. Go to Manage Ambari > Users
3. Reset the password for maria_dev to something secure.

It is a good idea to deactivate the other users or reset their passwords as well.

Clean up

To avoid any further charges, delete the instance:

- From the Compute Engine page select the instance and click on the DELETE button.

Note: A stopped instance is free, but its disk is not. The 50 GB disk costs about \$5/month.